Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 7

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

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Work done in Week 8

The results have been split up according to the various methodologies used in order to gain an understanding of what is going and not going so well. All the results are based on the DEV split, which contains 152 defect patches (representing 163 actual defects) and 280 defect-free patches. We have not made any changes to the TEST split data.

In this notebook, we have used 256×256 patches of AITEX dataset. It has 3222 defective patches and 4371 defect free patches. We split the data according to the source images. The defect-patches were split into DEV and TEST, and defect-free patches into REF (reference spectrum), CAL (normalisation and threshold) and NTEST (false alarms). Eight methods were tested, which included the LBP, LBP self-reference, Gabor, GLCM, three fixed Fourier pipelines (single-scale, pooled multi-scale and per-fabric multi-scale), and the adaptive multi-scale Fourier pipeline. In both methods we chose the threshold based on a Gumbel fit of the maximum score of the CAL patch with a desired false alarm rate of 5%. The same method of mask cleaning was followed for all methods.

Analyses:

- Fabric-wise results: Pixel AUROC, defect detection, well covered rate, image recall, false alarm rate, image AUROC were tested for each fabric. We also calculated the mean values. Fabrics of less than 5 images from the sources are indicated with a star.

- We also applied fixed multi-scale processing, each with their own set of references and thresholds, for each fabric. This was used to determine if the improvement was due to calibration or due to adaptive scale selection. We used paired bootstrap intervals to compare the results.

- We checked the Spearman correlation of weave period, peak strength and notch energy with the performance of each fabric.

- Defect-wise and size-wise results: We performed analysis of number of defects, median defect area, defect detection, defect coverage, defect recall, and pixel AUROC on different type and size of defects.

- We calculated bootstrap intervals for each method, group and metric by resampling the source images.

- Failure analysis: Defect area, contrast, changes in texture and distance from the patch edge. We also looked for any defects missed or discovered, and plotted the largest missed defect and the most confident false alarms.

- The key changes this week were the addition of a new control experiment (with the results reported separately for each fabric), and the inclusion of the LBP self-reference method. We saved the results in 22 CSV files, with names starting with w8_.

Results

All the results reported below are from the complete DEV run that includes 152 defect patches (163 actual defects, 280 defect-free patches). There are less than five source images for some fabrics and defect types. They mark their results with a star and ought to be considered observations rather than definite conclusions.

Overall comparison

The three Fourier pipelines with per-fabric calibration or adaptive scales detected a higher percentage of defects as compared to all the methods. The adaptive pipeline also performed the best at the pixel F1 score.

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============================================================
OVERALL RESULTS (DEV, 152 defect patches, 280 normal patches)
============================================================
                                 Pooled pixel F1  Defect detection  Image recall  False-alarm rate  Image AUROC
LBP                                        0.013             0.012         0.066             0.121        0.585
LBP self-reference                         0.026             0.067         0.105             0.111        0.790
GLCM                                       0.023             0.025         0.092             0.121        0.618
Gabor                                      0.022             0.080         0.158             0.121        0.755
FFT fixed single-scale                     0.023             0.252         0.257             0.018        0.783
FFT fixed multi-scale                      0.026             0.319         0.329             0.021        0.818
FFT fixed multi-scale (per-fabric)         0.049             0.485         0.507             0.039        0.838
FFT adaptive multi-scale                   0.091             0.503         0.500             0.018        0.819
```

The texture-based methods resulted in false alarms on approximately 11% of patches without defects, as compared to our 5% goal. The NTEST patches were not used to set their thresholds as they were also used to fit the texture baselines. These Fourier methods had fewer false alarms, ranging from 1.8% to 3.9%. However, they didn't do such a good job of matching the actual defects as evidenced by the low pixel F1 scores, and their masks also did not match up very well. The growth fraction that showed the best growth was 0.5, however, the growth index with the greatest growth was a positive 0.091, still on the edge of the search grid. The growth fraction was chosen on the same data used for F1 measurement (DEV) which could give a slightly optimistic result.

What is added to the adaptive method?

The improvement appears to be primarily due to calibration. The fixed pipeline also receives a very similar boost when using separate references and thresholds for each fabric. A change in the scales doesn't appear to provide much further improvement.
The overall effect of the scale-selection rule was similar to, but in the opposite direction, for macro pixel AUROC (−0.085). There was also no overall stretch that showed improvement, although it was slightly negative (−0.049). This may be due to the fact that the weave periods of all fabrics are very close (4.2 to 5.9 px). Because of our rule, which demands 4 weave periods, all fabrics are scaled (32, 64 and 128 px). The adaptive pipeline, therefore, is not all that different from the fixed pipeline, apart from this one-octave shift in scales and the change of sigma from 4.0 to 4.2–5.9.

By fabric

For not all fabrics this method gave the best result. The adaptive method was found to be good on Fabric 03, and the fixed method with different calibration for each fabric was better on Fabrics 02, 04 and 06.

FABRIC COUNTS (* = fewer than 5 source images)

fabric	source images	defect patches	true defects	normal patches
00	5	11	11	40
01*	4	8	8	40
02	18	33	36	40
03	13	85	91	40
04	6	10	11	40
05	0	0	0	40
06*	3	5	6	40

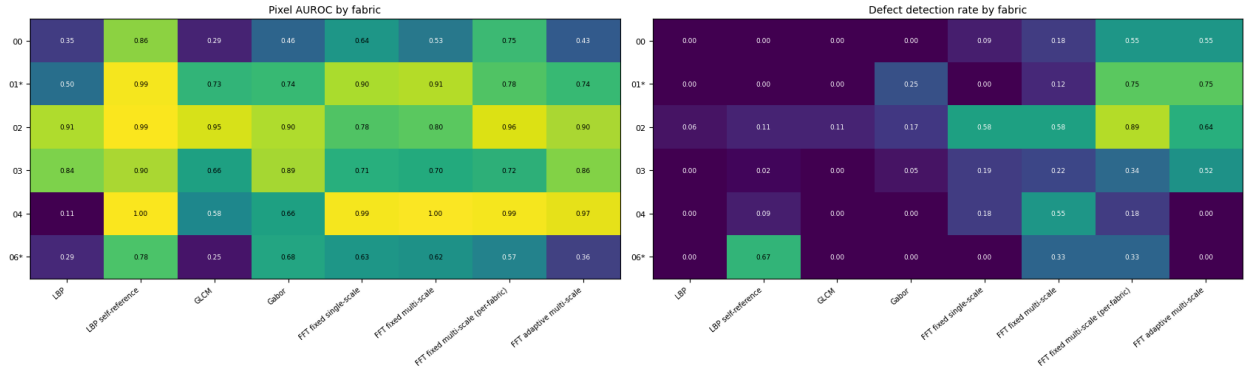
PIXEL AUROC BY FABRIC

	LBP	LBP self-reference	GLCM	Gabor	FFT fixed single-scale	FFT fixed multi-scale	FFT fixed multi-scale (per-fabric)	FFT adaptive multi-scale
00	0.346	0.865	0.285	0.460	0.635	0.528	0.748	0.432
01*	0.497	0.988	0.727	0.742	0.902	0.914	0.784	0.744
02	0.906	0.994	0.952	0.904	0.782	0.799	0.962	0.898
03	0.845	0.898	0.660	0.893	0.708	0.695	0.716	0.855
04	0.114	0.997	0.585	0.659	0.993	0.998	0.992	0.972
06*	0.293	0.779	0.248	0.682	0.630	0.617	0.567	0.355
mean over fabrics	0.500	0.920	0.576	0.723	0.775	0.758	0.795	0.710

DEFECT DETECTION RATE BY FABRIC (any predicted pixel touches the defect)

	LBP	LBP self-reference	GLCM	Gabor	FFT fixed single-scale	FFT fixed multi-scale	FFT fixed multi-scale (per-fabric)	FFT adaptive multi-scale
00	0.000	0.000	0.000	0.000	0.091	0.182	0.545	0.545
01*	0.000	0.000	0.000	0.250	0.000	0.125	0.750	0.750
02	0.056	0.111	0.111	0.167	0.583	0.583	0.889	0.639
03	0.000	0.022	0.000	0.055	0.187	0.220	0.341	0.516
04	0.000	0.091	0.000	0.000	0.182	0.545	0.182	0.000
06*	0.000	0.667	0.000	0.000	0.000	0.333	0.333	0.000
mean over fabrics	0.009	0.148	0.019	0.079	0.174	0.331	0.507	0.408

For Fabric 04, the adaptive method had an AUROC of 0.972 but did not detect any defects. This was because, even though defect scores were ranked higher than the normal pixel scores, the threshold (4.7) was still higher than the defect scores. The AUROC for Fabric 06 was < chance at 0.355. LBP self-reference yielded the highest macro pixel AUROC value (0.920), however, it only identified 14.8% of defects per fabric. This suggests that there are still problems with threshold selection. The average detection rate over fabrics was lower for the adaptive method than the fixed (per fabric) method (0.408 versus 0.507). But when we calculated the average across individual defects, the adaptive method was slightly better (0.503 compared with 0.485). This is primarily due to the fact that Fabric 03 contains 91 of the 163 defects; most of these defects were identified with the adaptive method.

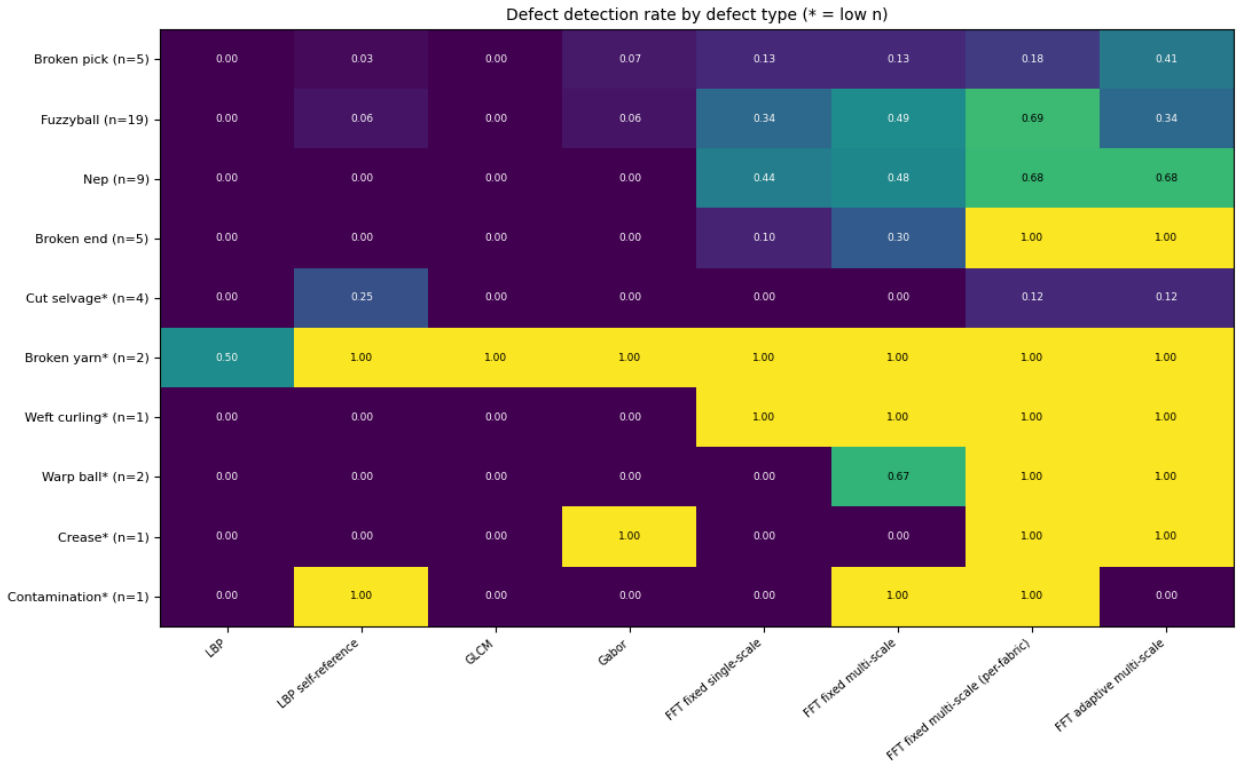


The difference between weave pattern and detection

We could not find a strong relationship between weave strength and detection performance. No correlation between items exceeded the Spearman correlation value of 0.8 which we used as a general guideline. The best performing values were −0.66 for period vs. fixed pixel AUROC and +0.66 for peak strength vs. Gabor pixel AUROC. Weave periods (4.2 to 5.9 px) and peak strengths (z = 63 to 71) are very similar for each of the fabrics. Due to this, it is hard to make a definite conclusion based on the correlation results.

By defect severity and size (AESI)

Out of the 163 defects, 106 belong to the two main defect types in this analysis: Broken pick (71 defects from 5 source images) and Fuzzyball (35 defects from 19 source images).
Both methods yielded valuable results for these two type of defect, but in different ways. The adaptive method outperformed the other methods on Broken pick (0.41 vs. 0.18). Fuzzyball was more covered than Broken pick (0.28). For small defect size, medium and large defect size the adaptive detection rate was nearly the same: 0.49, 0.50 and 0.52 respectively. For the fixed method, the detection decreased in all size groups (from 0.73 to 0.35). We did not have any Weft crack or Knot defects in the DEV split.

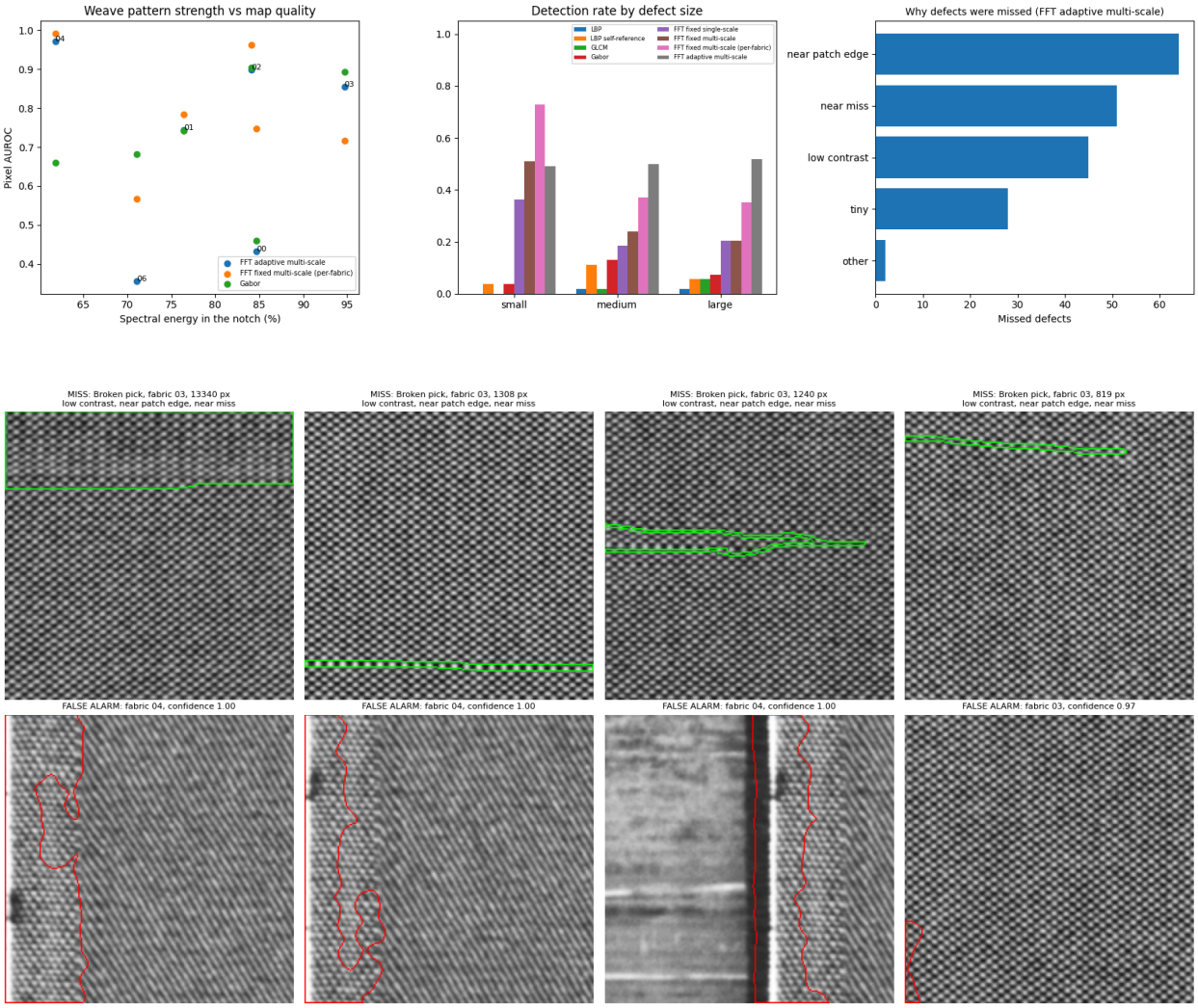


Failure analysis (adaptive pipeline)

81 of 163 defects were not detected by the adaptive pipeline. Of the defects that were missed, most were not too small but low in contrast.

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============================================================
MISSED VERSUS DETECTED DEFECTS (FFT adaptive multi-scale)
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           area  contrast  std_ratio  edge_dist  peak_ratio  defects
detected  433.0      0.84       0.91        0.0        1.37       82
missed    391.0      0.21       0.93        0.0        0.76       81
```

A tag could be attached to more than one defect. The tag counts were: near patch edge (64), near miss (51), low contrast (45), tiny (28) and other (2). The edge tag wasn't very helpful as the defects found were also at a similar distance from the patch edge. A near miss was counted when the score of a defect was 70% or more of the threshold. From this, 51 of the 81 defects that were not found (roughly 63%) were near missed defects. Fabrics 04 and 06 had 100% miss rate for defects, Cut selvage had 88% and Fuzzyball had 66%. The common characteristics of these tags suggest some similar defects that were missed, but do not provide open explanation for how they were missed.



Most of the false alarms on Fabric 04 were found near areas where texture and/or structure changes in a patch. These were not just the noises of randomness.

Confidence intervals

The overall detection rate was 0.50 for the adaptive method (95% CI: 0.41 to 0.60, 49 source images) and 0.48 for the per-fabric fixed method (95% CI: 0.36 to 0.69). The precision of the individual fabrics' confidence intervals was rather large. For instance, in one fabric, the adaptive detection rate was 0.55 and the interval was from 0.15 to 1.00, had been drawn only 5 source images. Intervals overlapped for most types of defects. The biggest discrepancy was between the Fourier methods and Gabor, with difference of less than 0.2 in most cases.

Challenges faced

- It was a challenge that some groups had very few source images. There are a total of 49 source images in the DEV split, with Fabric 01 being 4 and Fabric 06 being 3. Furthermore, 6 out of the 10 defect types have less than 5 images from which to source. Since we resampled source images and not individual patches for the bootstrap intervals, the results for these groups have wider intervals. The number of source images are indicated in each table and a star indicates groups of less than 5 images.

- However, Fabric 05 is not included in the result tables because there are no defects in the DEV split. Has 2 patches of defect in the TEST split.

- The comparisons between the fixed method and the adaptive method are rather unfair, since the parameters of the adaptive method (calibration and scale) were optimized with respect to the fixed method. The per-fabric fixed control and paired bootstrap results indicate that a significant amount of the improvement is due to calibration.

- We also examined the possibility of "bleeding" of patches in the different splits from the same source image. The splits were all done in the same manner – using the source images, with the same seed and draw order as Week 7.

- The thresholds for some methods did not give the expected results. Despite a target of only 5% accuracy, LBP, GLCM and Gabor were able to correctly identify only 11-12% of the normal patches. Another problem was that the adaptive method reached a pixel AUROC of 0.97 for Fabric 04, but it did not identify any defects. We have displayed both of these issues and the detection results, but they have not yet been resolved.

- Additionally, we experimented with a restriction in our examination of the relationship between weave properties and detection performance. With only six fabrics, it's too early to draw any conclusions. The correlation results are meant as representative.

- Results from the localisation scores could be higher than the actual performance. We selected the growth fraction using the same DEV data that we used for evaluation. The adaptive method choose 0.5 that is at the boundary point of the search grid. Thus, F1 score of 0.091 might be an optimistic score for the pixel F1.

- The tags indicate a class of defects that were missed, but do not indicate the actual reason behind the defect being missed. The 'near patch edge' tag was also not very useful as the defects were found within a similar distance from the edge.

- Some defects were always classified as 'object' (when testing with synthetic data). Due to this, a defect-wise analysis was restricted to one group. In addition, if two methods were found to be equivalent, the former method was considered the winner. Both these issues were corrected prior to the experiments with actual data.

- Time was also a factor to take into account. It took about 41-204 seconds to fit and 11-112 seconds to generate the maps for each method. We applied QUICK mode, set the number of reference, CAL and NTEST patches, and stored the deviation maps for the reference points in the cache to help with the repeated experiments.

Work To Be Done Next Week (Week 9): Fixed vs. Adaptive Comparison and Refinement

- This week's results will be used next week to make improvements to the methods. We will be discussing the defects that are common but which are not detected, which include defects that are very small, low-contrast defects and defects at the edge of a patch. We will also be discussing false alarms near patch corners and will review the types of defects that are difficult to detect.

- Only the development set will be used to make changes to the adaptive method. We will experiment with various window size rules using the number of weave repetitions over a window, various numbers of scales and different smoothing methods. We will also compare different ways of combining the scale maps, such as maximum, average and weighted sum.

- After making each of these main settings a single change at a time, we will see how much the results change. These include how strict the weave-peak detection is, how wide the frequency bands to remove are, which weave pattern to remove and what false-alarm rate to allow.

- The preresults of the preprocessing will be discussed and the Fourier method on the original patches will be compared to the same method on the preprocessing of the patches with CLAHE.

- In addition, we will attempt to limit the edge effects observed in this week's results, particularly the false alarms around patch corners and see if this makes a difference to the results.

- If we have time, another experiment will be conducted with the best Fourier method and the best baseline (descriptor fusion).

- The pipelines will be compared with the baseline methods: the fixed single-scale pipeline, fixed multi-scale and adaptive multi-scale pipeline. The comparison will include all metrics and the results for each fabric, defect type and defect size. Furthermore, an attempt will be made to uncouple the improvement due to per-fabric calibration from the improvement due to the adaptive scale rule. Confidence intervals will be included.

- Once the setting is complete, the methods will be executed on the TEST split that has not been used to tune the methods. We will then look at the TEST results in comparison with the DEV results and see if the methods have overfitted to the DEV data.

- Knowledge gained: Final tables and charts comparing methods, where the adaptive approach is successful and where it's not.

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